AWS Application Load Balancer

Understanding Path-Based and HTTP Method-Based Routing — A deep dive into intelligent Layer 7 traffic distribution for modern microservices architectures.

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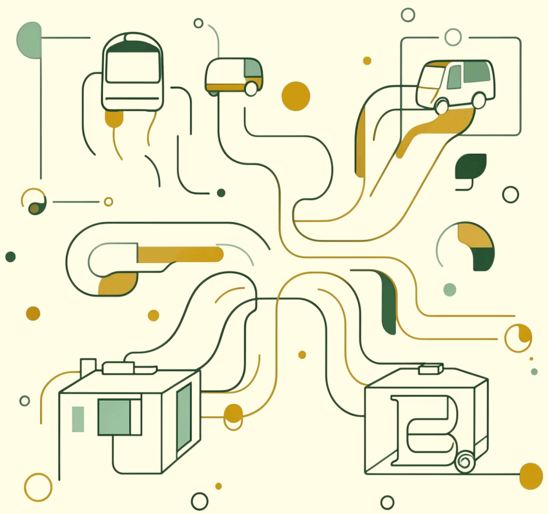
What Is the Application Load Balancer?



The AWS Application Load Balancer (ALB) operates at **Layer 7 — the Application Layer** of the OSI model. Unlike traditional load balancers that blindly distribute TCP connections, ALB **understands HTTP and HTTPS traffic**, inspecting every request before deciding where to send it.

Path Routing

Route by URL path

Method Routing

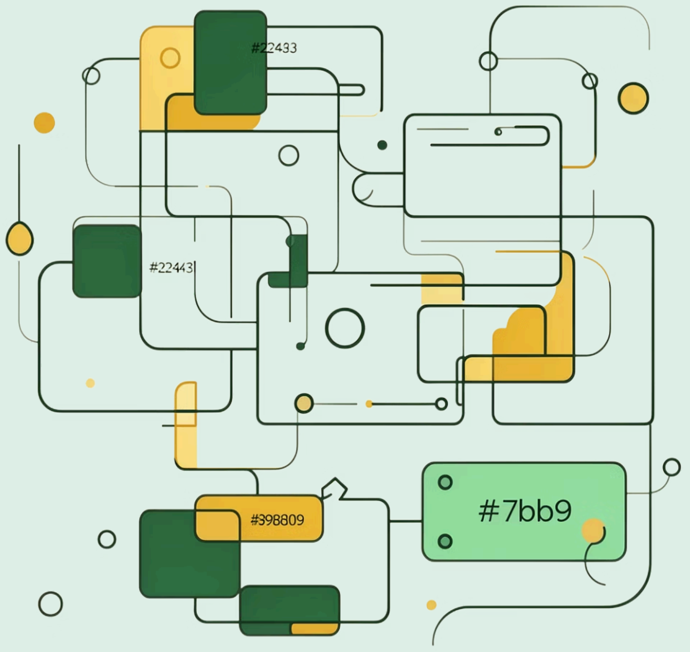
Route by HTTP verb

Host Routing

Route by hostname

Target Groups

Logical backend pools



Why ALB Matters in Microservices

Modern applications are decomposed into **independent services** — each owning a distinct business domain. ALB acts as the **single, intelligent entry point** that routes each request to the right service without clients needing to know the internal topology.

/products

Product catalogue service — independently scalable based on browse traffic.

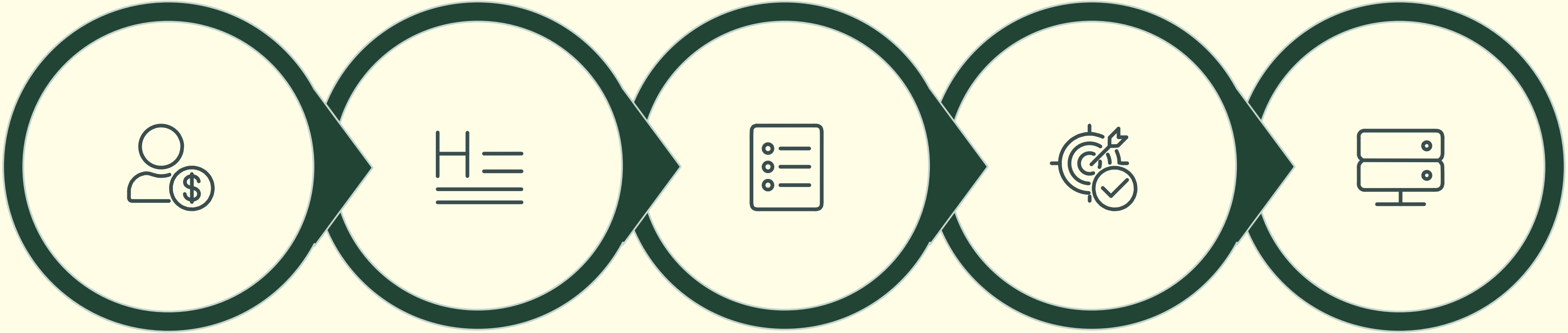
/cart

Shopping cart service — handles session-heavy read and write operations.

/payments

Payment service — isolated for security, compliance, and scaling needs.

How ALB Processes Every Request



Client Request

Inspect Headers

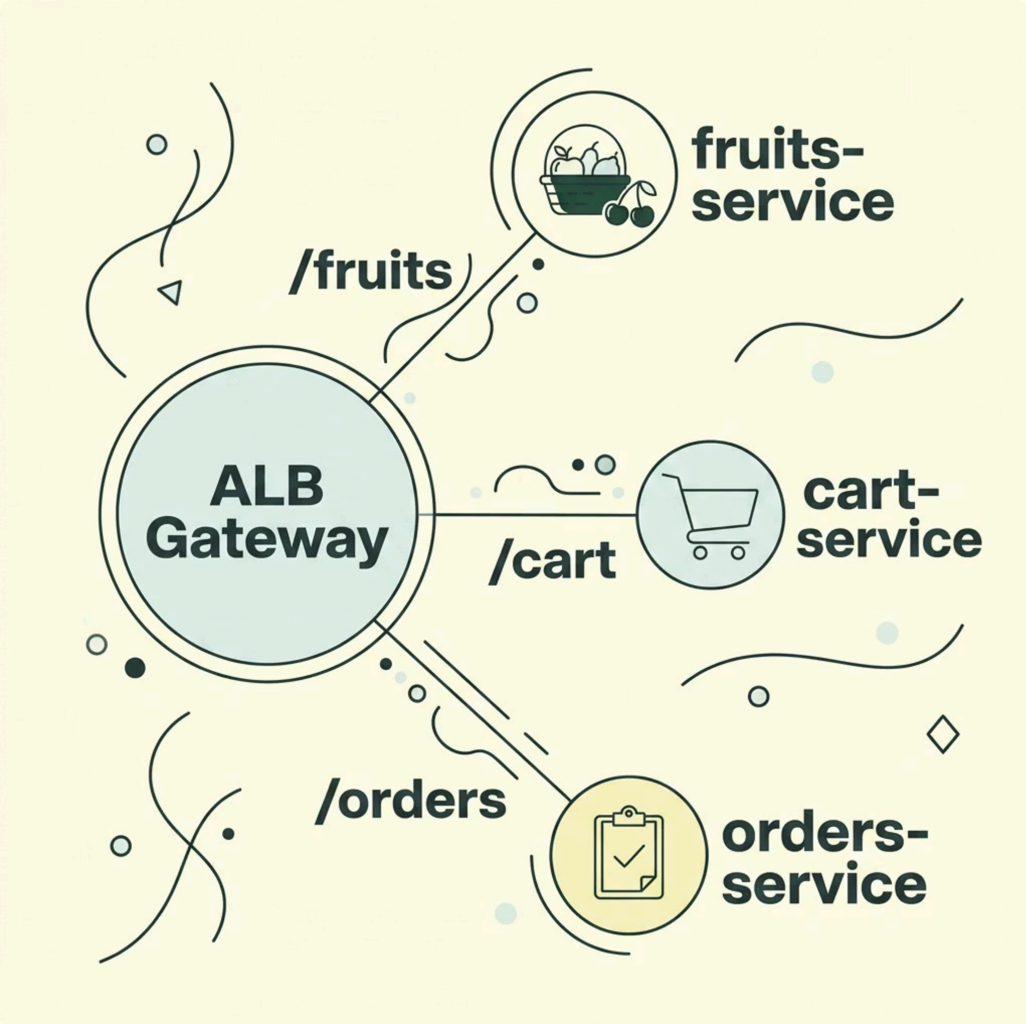
Evaluate Rules

Select Target

Forward to Backend

ALB evaluates requests against **ordered listener rules**, inspecting the path, HTTP method, headers, query strings, and hostname — the **first matching rule wins** and traffic is forwarded immediately to the corresponding target group.

Path-Based Routing Explained

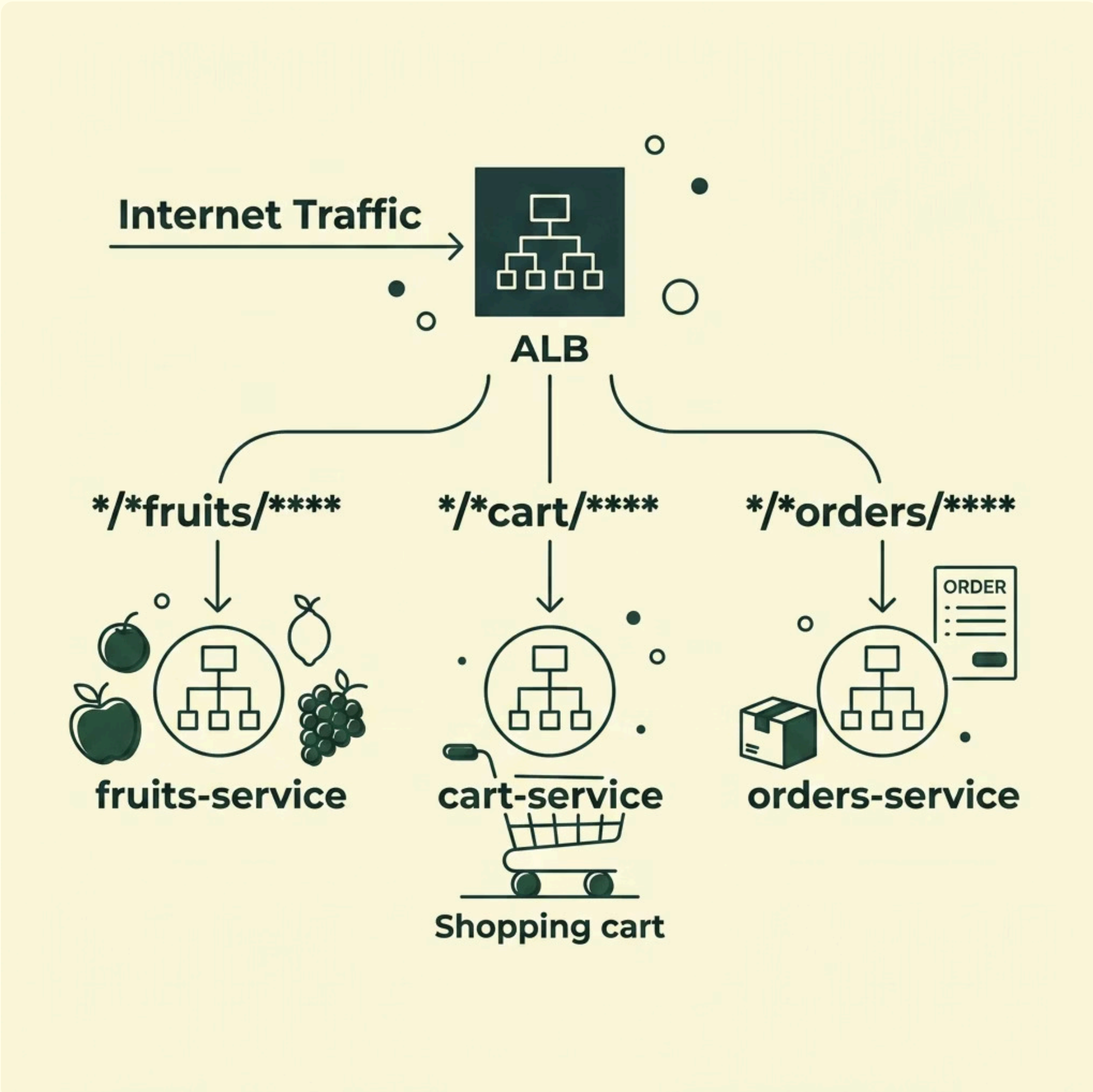


ALB inspects the **URL path** of every incoming HTTP request and forwards it to the service registered for that path. The path alone identifies *which microservice* should own the request.

> Example request:
>
> `GET /fruits/ HTTP/1.1`
>
> ALB matches `/fruits/*` → forwards to **fruits-service**.

- ***/fruits/**** → fruits-service target group
- ***/cart/**** → cart-service target group
- ***/orders/**** → orders-service target group

Real-World Path-Based Architecture



One ALB. Many Services.

A single ALB DNS endpoint serves the entire platform. Each path prefix maps to a dedicated backend — delivering clean **traffic separation** with no cross-service interference.

- Services scale independently per path
- Failures are isolated per service
- New services added without downtime
- Widely adopted microservices pattern

HTTP Method-Based Routing

Beyond the URL path, ALB can also **inspect the HTTP method** of each request. This means the same path can route to *different backend services* depending on whether the client is reading or writing data.

GET

Read / retrieve data from a service

POST

Create or submit new data to a service

PUT

Update or replace an existing resource

DELETE

Remove a resource from the service

Key principle: **Path identifies the service. HTTP method identifies the operation.** ALB evaluates both together when method-based rules are configured.

GET vs POST — Same Path, Different Backends

GET /cart/

Purpose: View the current cart contents

Routing: ALB → `cart-read-service`

- Read-only operation
- Optimised for high-frequency queries
- Can be cached or replicated

POST /cart/

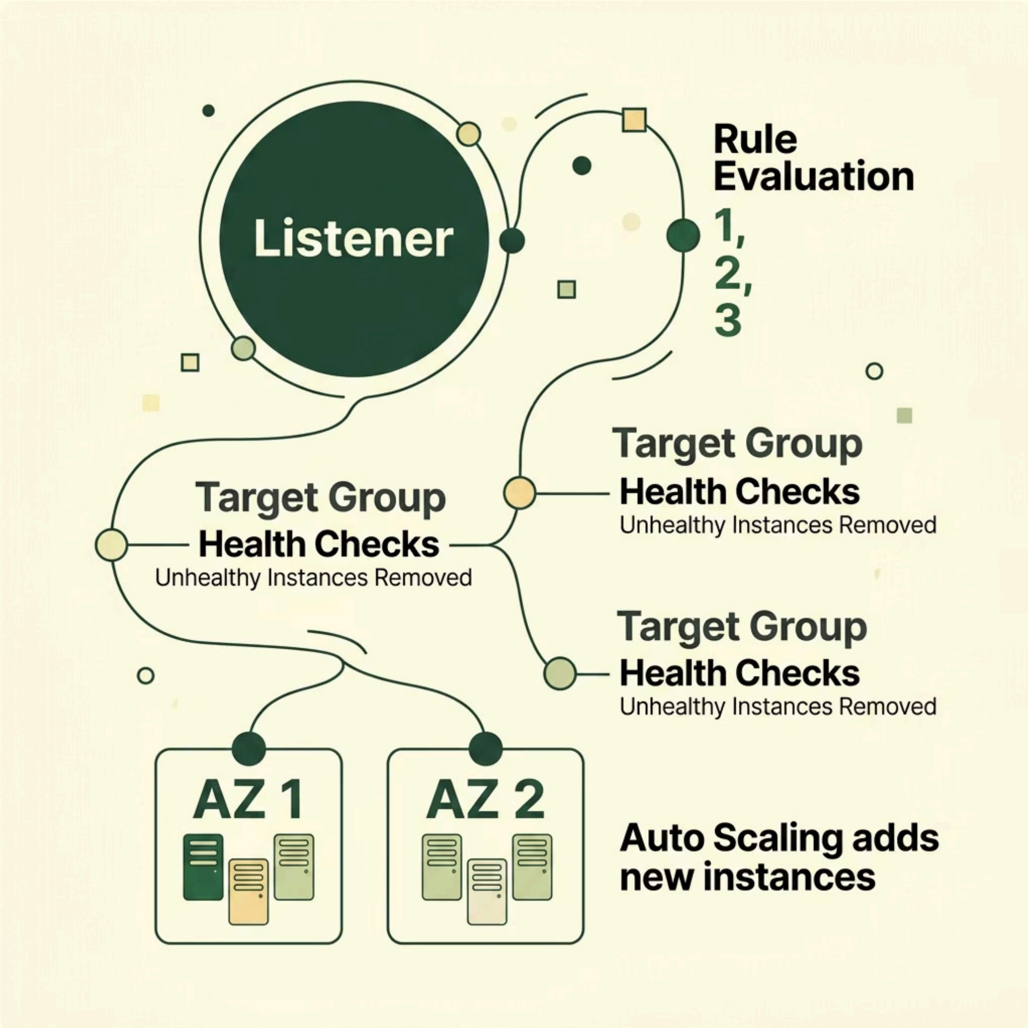
Purpose: Add a new item to the cart

Routing: ALB → `cart-write-service`

- Write operation with data mutation
- Requires stricter validation
- Isolated from read traffic

Listener Priority: The first matching rule wins. Method rules should be ordered *before* generic path rules to ensure correct routing.

Target Groups, Listener Rules & Scalability



How It All Fits Together

→ Target Groups

Logical pools of backend instances. Each microservice gets its own group.

→ Listener Rules

Ordered conditions (path, method, host) evaluated top-down. First match wins.

→ Health Checks

ALB continuously probes targets. Unhealthy instances are removed automatically.

→ Auto Scaling

Each target group scales independently based on its own load metrics.

Key Takeaways

Layer 7 Intelligence

ALB understands HTTP — it routes based on path, method, headers, and hostnames, not just IP addresses.

Path-Based Routing

URL paths map cleanly to microservices — `/products`, `/cart`, `/orders` each reach their own backend.

Method-Based Routing

The same path can route differently for GET vs POST — enabling read/write separation at the load balancer level.

Rules, Health & Scale

Priority-ordered listener rules, continuous health checks, and Auto Scaling ensure resilient, independent service operation.